

◆ 16.11.7 Decision and Control Optimality Integration (Final — Integrated Version)

Feasible Action Set

$$\mathcal{U}_{\{\text{feasible}\}} \subset \mathcal{U}, \quad |\mathcal{U}_{\{\text{feasible}\}}| < \infty$$

with bounded control operator $B(u)$. Thus:
an optimal action exists.

Optimal Decision Rule

$$u_k = \arg\max_{u \in \mathcal{U}_{\{\text{feasible}\}}} D_u(k)$$

Since the feasible set is finite and scores are bounded:

$$\exists u_k$$

Uniqueness (Tie-Breaking)

If multiple maximizers exist:

$$u^* = \arg\min_{u \in \text{ties}} C(u)$$

Thus:

a unique action is selected.

Decision Threshold (Safety Filter)

$$\text{execute if } D_u(k) > D_{\mathrm{crit}}$$

Otherwise:

$$u_k = 0$$

Thus:

weak or uncertain decisions are rejected.

Hysteresis (Anti-Oscillation Mechanism)

$$\text{switch if } D_{\mathrm{new}} > D_{\mathrm{old}} + \delta + \eta \sigma$$

Thus:

switching frequency is finite and oscillations are suppressed.

Decision Confidence

$$\mathrm{Conf}_k = D_{\{\max\}} - D_{\{\mathrm{second}\}}, \quad 0 \leq \mathrm{Conf}_k \leq 2C_D$$

Thus:

decision reliability is quantifiable.

Energy Consistency

$$\langle \nabla \mathcal{E}(\Psi_k), B(u_k) \rangle \leq 0$$

Thus:

control actions do not increase system energy.

Real-Time Guarantee

$$T_{\{\mathrm{decision}\}} = O(|\mathcal{U}_{\{\mathrm{feasible}\}}|)$$

Thus:

computation time is bounded and suitable for real-time operation.

Closed-Loop Injection

$$\begin{aligned} \Psi_{k+1} = \\ \Psi_k + dt \cdot F(\Psi_k, u_k) \end{aligned}$$

Thus:

decisions directly influence system evolution.

Convergence and Deterministic Limit

$$\limsup_{k \rightarrow \infty} D(k) = D^*$$

$\lim_{k \rightarrow \infty} u_k = u^*$

$\lim_{\sigma \rightarrow 0} \lim_{k \rightarrow \infty} u_k = u^*$

Thus:

the decision process converges to an optimal policy.

Stability Compatibility

Since:

- $D_u(k)$ is bounded
- switching is finite (hysteresis)
- control respects energy dissipation

Thus:

the decision layer preserves Lyapunov stability.

Implication

$\boxed{\text{The decision mechanism is optimal, unique, stable, and physically consistent within the}}$

closed-loop system.}
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**SEKHEM-TSDD Theory, Mohamed Abd Elnaby,
DOI: 10.5281/zenodo.19521219"**